

J. SARGEANT REYNOLDS COMMUNITY COLLEGE

FLUORESCENT PROTEIN ENGINEERING BOOTCAMP

LABORATORY SAFETY AGREEMENT

The laboratory is an integral component of your training in the Fluorescent Protein Engineering Bootcamp at J. Sargeant Reynolds Community College. This is a Biosafety Level 1 (BSL-1) molecular biology laboratory in which you will culture baker's yeast (*Saccharomyces cerevisiae*), handle chemical reagents, perform gene-editing and cloning procedures, and visualize fluorescent proteins. When the rules below are followed, the lab is a safe learning environment. Read these rules carefully, then sign and date the agreement on the last page. By signing, you agree to abide by the rules of the laboratory. Only after this agreement is signed will you be allowed to work in the lab.

DRESS CODE

- ▶ Wear clothing that, if damaged or stained by reagents or dyes, would not be a serious loss.
- ▶ Wear a buttoned lab coat at the bench when instructed, and wear closed-toe shoes at all times. **No open-toed shoes, flip-flops, or sandals are allowed in lab.**
- ▶ Tie back long hair, loose clothing, and dangling jewelry while in the laboratory, especially near Bunsen burners and open flames.
- ▶ Wear safety goggles and gloves when working with chemicals, hot media, glassware, or staining reagents. Change gloves whenever they are contaminated.

PERSONAL CONDUCT

- ▶ Horseplay and pranks are never appropriate in the lab.
- ▶ Do not touch any equipment, reagents, or cultures unless instructed.
- ▶ **Do not eat, drink, chew gum, apply cosmetics, or handle contact lenses in the lab.** Never eat or drink from lab glassware, and never store food or beverages in lab refrigerators or freezers.
- ▶ Do not taste, smell, or directly touch any chemical, medium, or yeast culture.
- ▶ Wash your hands with soap and water before leaving the lab, even if gloves were worn.

WORKING WITH YEAST & BIOLOGICAL CULTURES

- ▶ Treat all microorganisms, including baker's yeast, as if they were capable of causing harm. Use aseptic technique at all times when working with cultures, plates, and media.

- ▶ The yeast strains and engineered plasmids used in this lab are non-pathogenic BSL-1 organisms. Even so, all cultures and genetically modified material must stay in the lab and be handled and disposed of as directed.
- ▶ Disinfect your bench with 70% ethanol or the provided disinfectant before and after each session.
- ▶ Work near a flame or as directed when opening culture tubes and plates to reduce contamination. Keep lids closed when not in active use.
- ▶ **Place all used plates, tubes, tips, and culture-contaminated items into the designated biohazard / autoclave waste container.** Never place biological waste in the regular trash or down the drain.
- ▶ Report any spill of culture or genetically modified material to your instructor immediately so it can be properly decontaminated.

CHEMICALS & REAGENTS

- ▶ This lab uses reagents including lithium acetate, PEG, ethanol, agarose, DNA stains, and antibiotic/selection agents. Treat every reagent as potentially hazardous and read the label before use.
- ▶ To avoid contamination, take only what you need and do not return unused reagent to the stock container.
- ▶ **Use a micropipette and the appropriate tip for measuring small volumes. Never pipette by mouth.**
- ▶ Handle DNA stains (such as gel stains) only with gloves, and dispose of stained gels and buffers only in the labeled waste container.
- ▶ When diluting acids, always add acid to water — never add water to acid.
- ▶ Dispose of chemicals only as directed. Do not pour chemicals down the drain or into the trash unless your instructor tells you to.
- ▶ If any chemical contacts your skin or eyes, flush with copious amounts of water immediately and notify your instructor.

EQUIPMENT & INSTRUMENTS

- ▶ Handle micropipettes, centrifuges, vortexers, thermocyclers, and water baths only as directed. Never exceed a micropipette's volume range or rotate the dial past its limits.
- ▶ Balance the centrifuge before every run and never open the lid until the rotor has fully stopped.
- ▶ Use extreme care with heat sources, including Bunsen burners, hot plates, heat blocks, and water baths. Never point the open end of a heated tube toward yourself or anyone else.

- ▶ **When using the fluorescence imager, transilluminator, or any UV or blue-light source, use the provided shield, face protection, and skin coverage.** Never expose skin or unprotected eyes to UV light.
- ▶ Operate the gel electrophoresis power supply only with the lid in place; high voltage is present. Disconnect power before opening the chamber.
- ▶ Handle sharp items — scalpels, glass, and slides — with care, and place broken glass and sharps only in the designated sharps container.

PERFORMING AN EXPERIMENT

- ▶ Keep your work area clear of anything not needed for the procedure. Stow bags, coats, and books away from the bench.
- ▶ Monitor your experiment personally at all times; never leave a running instrument or open flame unattended.
- ▶ Do not work alone in the lab. Always work with a partner or under supervision unless directed otherwise.
- ▶ Report all spills, broken glass, or missing or damaged equipment to your instructor immediately.
- ▶ When finished, turn off all equipment, decontaminate and clean your bench, rinse glassware as directed, and return all materials to their proper place.
- ▶ If anything is unclear, ask your instructor before proceeding.

FIRST AID & EMERGENCIES

- ▶ Report any incident, accident, exposure, or injury to your instructor immediately, no matter how minor it seems.
- ▶ Inform your instructor of any chemical or biological contact and wash the affected area with copious amounts of water.
- ▶ Know the locations of the safety equipment in your laboratory: fire extinguisher, eyewash station, safety shower, first aid kit, spill kit, and biohazard waste. Learn how to use each.
- ▶ Know the shortest exit route from the lab and the building, and the location of the nearest telephone in case 911 must be dialed.

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LABORATORY SAFETY AGREEMENT**

- ▶ I have read and agree to follow all of the safety guidelines set forth in this agreement, including the procedures for handling yeast cultures and genetically modified material.
- ▶ My instructor has informed me of the locations of fire safety equipment, the eyewash station, the spill kit, and biohazard waste containers.
- ▶ I realize that I must obey these rules to ensure my own safety and that of my fellow students and instructors. I shall cooperate fully and follow closely the oral and written instructions provided by the instructor.
- ▶ I am aware that any violation of this safety agreement that results in unsafe conduct or misbehavior may result in being removed from the laboratory, a failing grade, and/or dismissal from the bootcamp without refund.
- ▶ I am aware that wearing open-toed shoes means I will not be allowed to remain in lab or participate in lab activities, and I may receive a zero for that session.
- ▶ If I have a medical condition (for example, pregnancy, allergies, latex sensitivity, or a condition affected by UV exposure) or am taking medication (for example, immunosuppressive drugs), I will discuss the safety of my participation in this laboratory with my physician.

Student, print name _____

Student signature _____

Date _____

Parent/Guardian, print name *(if under 18)* _____

Parent/Guardian signature *(if under 18)* _____

Date _____

In case an emergency situation arises, please contact:

(name)	(phone number)
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Instructor's Signature _____

Date _____